



EAST CHINA NORMAL UNIVERSITY
SCHOOL OF MATHEMATICAL SCIENCES

Mathematics Syllabus

Summer Seminar on Galois Theory

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Semester: Summer vacation, 2026–2027
Classroom: Fourth Teaching Building, Room 211
Time: 18:00–20:00, daily (Mon–Fri)
Prerequisites: Abstract Algebra (I)

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Course Description

This seminar offers an intensive introduction to Galois Theory, one of the most elegant and profound achievements in modern algebra. Beginning with a rapid review of field extensions and algebraic closure, we will develop the core correspondence between subgroups of the Galois group and intermediate fields, establishing the fundamental theorem in full generality. We will then explore solvability by radicals, the insolubility of the general quintic, and the computation of Galois groups for polynomials of low degree. Depending on time and participant interest, we may touch upon infinite Galois theory, Kummer theory, or applications to algebraic number theory.

Learning Objectives

By the end of this seminar, participants will be able to:

- (1) Understand the Galois correspondence and the Fundamental Theorem of Galois Theory, and apply it to analyze the structure of field extensions.
- (2) Compute Galois groups for polynomials of degree up to 5, and determine whether a given polynomial is solvable by radicals.
- (3) Master key techniques including splitting fields, normal and separable extensions, and the discriminant criterion.
- (4) Develop proficiency in reading and writing rigorous algebraic proofs at an intermediate graduate level.
- (5) (Optional / Advanced) Be introduced to infinite Galois theory and its connections to algebraic number theory and inverse Galois problems.

Course Content and Schedule

Part I: Field Extensions and Splitting Fields (4 contact hours)

- Field extensions: degree, algebraic and transcendental elements, minimal polynomials, simple extensions, and the tower law.
- Existence and uniqueness of algebraic closures; construction and universal property of splitting fields; normal extensions and equivalent characterizations.

Part II: Separability and the Galois Correspondence (6 contact hours)

- Separable extensions: separable degree, perfect fields, and the primitive element theorem.
- Definition of the Galois group; fixed fields and the fixed-field correspondence.
- The Fundamental Theorem of Galois Theory: proof of the Galois correspondence, index equals degree, and normality corresponds to normality.
- Worked examples: Galois groups of finite fields and cyclotomic extensions.

Part III: Solvability by Radicals (4 contact hours)

- Solvable groups: derived series, composition series, and equivalent conditions for solvability.

- Radical extensions and solvable extensions; Galois's criterion for solvability by radicals.
- The general polynomial of degree n and its Galois group over the function field; **the insolvability of the general quintic** (Abel–Ruffini theorem).

Part IV: Computing Galois Groups (4 contact hours)

- Discriminant as a Galois resolvent; Galois groups of cubic polynomials and the sign of the discriminant.
- Quartic polynomials: resolvent cubics, the dihedral and alternating cases, and explicit criteria for S_4 versus D_4 .
- Reduction modulo p and Dedekind's theorem; applying cycle types to determine Galois groups of quintic polynomials.
- Summary of algorithms and computer algebra approaches.

Part V: Advanced Topics (2 contact hours)

- Infinite Galois theory: Krull topology on the absolute Galois group, profinite groups, and the infinite Galois correspondence; or Kummer theory and abelian extensions (time-permitting).

Assessment Methods

This seminar employs process-oriented assessment with no final written examination.

Textbook

李文威. 代数学方法 (第一卷). 高等教育出版社, 2019.

References

- [1] Emil Artin. Galois Theory. Dover Publications, 1998.
- [2] Ian Stewart. Galois Theory. 4th ed. CRC Press, 2015.
- [3] David A. Cox. Galois Theory. 2nd ed. John Wiley & Sons, 2012.
- [4] Michael Artin. Algebra. 2nd ed. Pearson, 2010.
- [5] Serge Lang. Algebra. Rev. 3rd ed. Springer, 2002. (GTM 211)